

Methods to solve non-linear Equations

1) Bisection Method (Bracketing method)

Used when the function changes sign over an interval $[a, b]$, $f(a) * f(b) < 0$.

Choose x_0 and x_1 such that $f(x_0) * f(x_1) < 0$.

Compute mid-point $x_2 = (x_0 + x_1) / 2$.

Check sign of $f(x_2)$. Replace x_0 or x_1 accordingly.

Advantage: Always converges.

Disadvantage: Slow convergence.

2) False Position Method (Regula Falsi)

Similar to Bisection, but uses a weighted average:

$$x = (x_0 * f(x_1) - x_1 * f(x_0)) / (f(x_1) - f(x_0))$$

Better approximation than mid-point.

Advantage: Faster than bisection.

Disadvantage: Can get stuck if the slope doesn't change quickly.

Partial Differentiation & Euler's Theorem

Note: If $u(x, y) = f(x, y) + g(x, y) + h(x, y)$, where f , g , and h are homogeneous functions with degree m , n , and p respectively,

$$(a) \quad x \frac{du}{dx} + y \frac{du}{dy} = m f + n g + p h$$

$$(b) \quad x^2 \frac{d^2u}{dx^2} + 2xy \frac{d^2u}{dxdy} + y^2 \frac{d^2u}{dy^2} = m(m-1)f + n(n-1)g + p(p-1)h$$

If $f(u)$ is a homogeneous function in two variables x & y with degree n , then:

$$(a) \quad x \frac{du}{dx} + y \frac{du}{dy} = n \cdot f(u) / f'(u) = F(u)$$

$$(b) \quad x^2 \frac{d^2u}{dx^2} + 2xy \frac{d^2u}{dxdy} + y^2 \frac{d^2u}{dy^2} = F(u) \cdot [F'(u) - 1]$$

Example: If $u = \arctan(y/x)$, show that:

$$(i) \quad x \frac{du}{dx} + y \frac{du}{dy} = 0$$

$$(ii) \quad x^2 \frac{d^2u}{dx^2} + 2xy \frac{d^2u}{dxdy} + y^2 \frac{d^2u}{dy^2} = 0$$

Solution:

$$\frac{du}{dx} = \left(\frac{1}{1 + (y/x)^2} \right) \cdot \left(-\frac{y}{x^2} \right) = -\frac{y}{x^2 + y^2}$$

$$\frac{du}{dy} = \left(\frac{1}{1 + (y/x)^2} \right) \cdot \left(\frac{1}{x} \right) = \frac{x}{x^2 + y^2}$$

$$\text{Therefore, } x \frac{du}{dx} + y \frac{du}{dy} = x \cdot \left(-\frac{y}{x^2 + y^2} \right) + y \cdot \left(\frac{x}{x^2 + y^2} \right) = 0.$$

Differentiating partially w.r.t x and y :

$$\frac{du}{dx} + x \frac{d^2u}{dx^2} + y \frac{d^2u}{dxdy} = 0$$

$$x \frac{d^2u}{dxdy} + \frac{du}{dy} + y \frac{d^2u}{dy^2} = 0$$

Multiplying first by x , second by y , and adding:

$$x \frac{du}{dx} + y \frac{du}{dy} + x^2 \frac{d^2u}{dx^2} + 2xy \frac{d^2u}{dxdy} + y^2 \frac{d^2u}{dy^2} = 0$$

Since $x \frac{du}{dx} + y \frac{du}{dy} = 0$:

$$\text{Hence, } x^2 \frac{d^2u}{dx^2} + 2xy \frac{d^2u}{dxdy} + y^2 \frac{d^2u}{dy^2} = 0.$$